

Lab Question Based on While Loop

1. Write a program to find the largest digit in a number using a while loop.

```
num = int(input("Enter a number: "))
max_digit = 0
while num > 0:
    digit = num % 10
    if digit > max_digit:
        max_digit = digit
    num //= 10
print("Largest digit is:", max_digit)
```

2. Write a program to print the Fibonacci series up to n terms using a while loop.

```
n = int(input("Enter number of terms: "))
a, b = 0, 1
count = 0
while count < n:
    print(a, end=" ")
    a, b = b, a + b
    count += 1
```

3. Write a program to check whether a number is an Armstrong number using a while loop.

□ *Armstrong number*: A number is Armstrong if the sum of cubes of its digits equals the number itself (e.g., $153 \rightarrow 1^3 + 5^3 + 3^3 = 153$)

```
num = int(input("Enter a number: "))
original = num
sum = 0
while num > 0:
    digit = num % 10
    sum += digit ** 3
    num //= 10
if sum == original:
    print("Armstrong number")
else:
    print("Not an Armstrong number")
```

4. Keep summing the digits of a number until a single-digit number is obtained (e.g., $9875 \rightarrow 9+8+7+5=29 \rightarrow 2+9=11 \rightarrow 1+1=2$)

```
num = int(input("Enter a number: "))
while num >= 10:
    temp = num
    sum_digits = 0
    while temp > 0:
        sum_digits += temp % 10
        temp //= 10
    num = sum_digits
```

```
print("Single digit sum:", num)
```

5. Check whether a number is a perfect number (e.g., $28 \rightarrow 1 + 2 + 4 + 7 + 14 = 28$)

```
num = int(input("Enter a number: "))
i = 1
sum_divisors = 0
while i < num:
    if num % i == 0:
        sum_divisors += i
    i += 1
if sum_divisors == num:
    print("Perfect number")
else:
    print("Not a perfect number")
```

6. Find the GCD (Greatest Common Divisor) of two numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
while b != 0:
    a, b = b, a % b
print("GCD is:", a)
```

7. Program to reverse order of words.

```
s=input("Enter Some String:")
l=s.split()
l1=[]
i=len(l)-1
while i>=0:
    l1.append(l[i])
    i=i-1
output=' '.join(l1)
print(output)
```

8. Program to reverse internal content of each word.

```
s=input("Enter Some String:")
l=s.split()
l1=[]
i=0
while i<len(l):
    l1.append(l[i][::-1])
    i=i+1
output=' '.join(l1)
print(output)
```

9. Write a program to remove duplicate characters from the given input string?

```
s=input("Enter Some String:")
l=[]
for x in s:
    if x not in l:
        l.append(x)
output="".join(l)
print(output)
```

10. Write a program to find the number of occurrences of each character present in the given String?

```
s=input("Enter the Some String:")
d={}
for x in s:
    if x in d.keys():
        d[x]=d[x]+1
    else:
        d[x]=1
for k,v in d.items():
    print("{} = {} Times".format(k,v))
```